ES expansion and differentiation into myoblasts

Phase 1: mESC expansion (Proliferation & maintaining pluripotency)

ESC thawing and culture

- Thaw ESCs (1E6 cells aliquots) by letting the tube warm up ca. 5min at RT until only a small ice clump is left in the center of the cell suspension in freezing medium. For EB cultures, thaw cells 2 days in advance.
- Add 500ul of DMEM/10%FBS/Pen+Strep, transfer into 10ml of DMEM/10%FBS/Pen+Strep in a 15ml tube.
- Centrifuge 4min at 260g, take up in 4ml 2i/LIF ESC medium.
- Plate on 1 well in 6w dish pre-coated with LN521.
- Passage ES cells every 2-3 days 1:4 to 1:8. Change medium daily.
- Freeze ES cells in 500µl/1E6 aliquots in Recovery Freezing medium (Gibco 12648010).

Reagents: 2i/LIF ESC medium

To prepare 50 ml of medium:
23.5ml DMEM/F12 medium (Gibco 11320033)
23.5ml Neurobasal medium (Gibco 21103049)
0.5ml 100x Pen+Strep (Gibco 15140122)
0.5ml 200mM L-Glutamine (Gibco 25030081)
50µl of 1000x (55mM) 2-mercaptoethanol (Gibco 21985023)
0.5ml 50x Neurobrew-21 w/o vitamin A supplement (Miltenyi 130-097-263)
0.25ml 100x N-2 supplement (Gibco 17502001)
5µl PD0325901 (Bio-Techne 4192/10) 10mM stock in DMSO; working concentration: 1µM
15µl CHIR99021 (Bio-Techne 4423/10) 10mM stock in DMSO; working concentration: 3µM
10µl 25mg/ml Primocin (InVivoGen)
3.33 uL of Human LIF (stock is 0.15mg/mL)
1ml ESC-qualified FBS (Gibco 16141061)
1 ml 5% sterile BSA - dissolve BSA fraction V (Roche 10735086001) in PBS (Sigma D8537-500ML), then sterile filter solution with 0.22µm membrane.

Pre-coating of culture plates

- Dilute laminin-521 (BioLamina LN521) in PBS +Ca2+ to a working concentration of 0.5µg/ml and add to plate (6w plate: 1.5 ml per well) (7.5ul in 1.5ml PBS +Ca2+).
- Incubate plates O/N in fridge (4-8C). Alternative: incubate plates 2h at 37C in the incubator.

Phase 2: EB formulation (Differentiation)

Day -1

Coat 96-well suspension plate with Lipidure (Lipidure-CM5206) and let evaporate overnight in hood
Lipidure (95% in Ethanol), dilute 1:10 (in our case 1:5 due to stock dilution) in 100% Ethanol and add 100ul per well.

Day 0 to 2

Colonies dissociated with trypsin and re-plated in suspension dishes in myoblast medium to become EBs.

Plate cells into Lipidure coated U-bottom 96-well plate, 3E4 cells/well in 100 µl Myoblast medium. Spin down 4 min at 220g and culture for 2 days.

- Harvest EBs from 96wp with multi-channel pipette, transfer to 10 cm dish.
- Collect EB suspension in 15 ml tube, flush dish once with an additional 5 ml DMEM.
- Let EBs sediment for 3-4 min, remove supernatant.

Myoblast medium
36.5 ml IMDM with glutamine (Gibco)
7.5 ml (15%) FBS
5 ml (10%) horse serum
0.5 ml (1%) Chick Embryonic Extract
50 µl (1:1000) of 50 mg/ml stock ascorbic acid
20 µl (1:2500) of 411.5M (97%) monothioglycerol
0.5 ml (1:100) Pen +Strep (Gibco)
100 µl Primocin

2.1 Inducing myogenesis

Pre-requisite: Matrigel coated T25 flasks/8well plate

Day 2 to 5
Transfer EBs transferred to Matrigel coated T25 flasks/8well plate(imaging) in Myoblast medium supplemented with **2uM (given in lab 10 mM) doxycycline** and **10uM SB431542 (1:1000).** Doxycycline to activate the tetracycline response element (TRE), inducing expression of MyoD.
- Cultured for 3 days.
- Coat plates with Growth Factor reduced Matrigel (1:50) in DMEM.
If EBs are put into IBIDI 8well u-slide, add 4-5 EBs per well. Add Doxycycline 1:1000. If you want to enhance myogenesis with SB (10uM end conc.) (TGF beta inhibitor), start from Day 2-5.
*Observe if cells are already differentiated, and are collapsing, you can fix them at Day 5 prior the myofiber media.

Day 5
Dissociate cells with AccuMAX, plate iMyoD MPCs on 8-well slides coated with GFR-Matrigel at 1-5 x 105 per well in 0.4mL myoblast medium(as sudden transformation after dissociation is a stress also).
Comment: At this point, the MPCs are re-plated for the last time. Alternative culture formats include nanofiber-scaffolds in 96w-sized wells – 4E4 cells per well

Phase 3: Maturation

Day 6
Myoblast medium replaced with Myofibre medium. Replaced every 2 days.

Myofibre Medium - Reagents
DMEM (low Glucose)
1% Horse Serum
1% Pen/Strep
1% B27 without insulin
100 ul Primocin

Notes:
T**ransitioning from 3D to 2D Monolayer:** EBs are three-dimensional clusters of cells. Dissociating them with AccuMAX on Day 5 breaks these aggregates apart. This allows the cells to be laid out flat as a single layer (monolayer) on slides or scaffolds. **Controlling Cell Density:** Breaking the EBs into individual cells allows scientists to accurately count and plate them at a highly specific density ($1–5 \times 10^5$ cells per well on slides, or 4×10^4 cells on nanofiber-scaffolds). Consistent density is critical for uniform cell behavior and muscle development. Once the EBs are dissociated and their physical structure is lost on Day 5, the protocol explicitly identifies these cells as: **iMyoD MPCs** (Myogenic Progenitor Cells).